Impact-Time-Control Guidance Law for Anti-Ship Missiles

In-Soo Jeon, Jin-Ik Lee, and Min-Jea Tahk

***Abstract*—In this paper, a new guidance problem with the impact time constraint is investigated, which can be applied to salvo attack of anti-ship missiles. The closed form solution based on the linear formulation is derived, suggesting an additional loop for adjusting the impact time in addition to the traditional optimal guidance loop. This solution is a combination of the well-known PNG law and the feedback of the impact time error, which is the difference between the impact time by PNG and the prescribed impact time. The new guidance law called ITCG (Impact-Time-Control Guidance) can be used to guide multiple missiles to hit a stationary target simultaneously at a desirable impact time. Nonlinear simulation of several engagement situations demonstrates the performance and feasibility of ITCG. In addition, the similarity of the closed form solution and APNG is investigated and the switching rule for practical implementation is discussed.**

***Index Terms*—Anti-ship missile, impact-time-control, minimum energy control, missile guidance, optimal control.**

I. Introduction

MANY advanced guidance laws have been devised for decades either to improve the guidance performance of the conventional PNG (Proportional Navigation Guidance) law or to achieve some specific objectives which can not be attained otherwise [1]–[11]. Minimum time, minimum control effort, or prescribed impact angle achievement are good examples of those objectives. The shorter the engagement time, the better the missile survives the threats, as the reaction time of the target against the missile is reduced. Minimum control effort strategies allow the missile to react readily against uncertain disturbances in the final stage. Also, to insure a high kill probability, the achievement of a proper attitude angle is desirable for some weapon systems, e.g. anti-tank missiles, torpedos, or anti-ship missiles [1], [2]. In this case, of course, the terminal impact angle deserves to be controlled at the expense of control efforts. Another important objective or constraint in missile homing guidance is flight time. Main concerns on flight time have been focused on minimum time homing [3], precise time-to-go estimation [4], sensitivity reduction of time-to-go estimates [5], or homing techniques without time-to-go estimation [6]. In this work, the guidance problem with a new kind of constraint on the impact time will be addressed, which has not attracted much attention so far. Our interest in flight time will be changed from minimum flight time to meeting an arbitrarily designated flight time.

Modern warfare ships have a number of self-defense measures against anti-ship missiles, such as surface-to-air missile systems, ECM systems, and CIWS (close-in weapon system). CIWS is a naval shipboard weapon system for detecting and destroying incoming anti-ship missiles and enemy aircraft at short range. Nearly all classes of modern warships are equiped with CIWS system. CIWS usually consists of a combination of radars, fire-control-systems, and multiple rapid-fire guns. It has been a great obstacle of anti-ship missiles to complete their missions. Salvo attack of anti-ship missiles, in which several missiles hit the target as *simultaneously* as possible, has been devised as one of countermeasures to survive the threat of CIWS, taking advantage of the vulnerability of CIWS's one-by-one engagement feature. A representative way is a pre-programmed midcourse salvo strategy in which the missiles, launched from different platforms or launched consecutively from a single platform, approach to the expected target position via various waypoints; however, unpredictable target movement during the midcourse guidance of the missiles before target acquisition of the seeker makes it difficult to achieve simultaneous attacks on the target. In this work, a feedback guidance law to hit the target at the designated impact time is considered as an alternative way to cope with this drawback of the midcourse salvo strategy. Of course, the designated impact time is determined in consideration of the engagement situation. The homing guidance law for this aim will be called an impact-time-control guidance law in this work.

The well-known PNG law is optimal for a constant-velocity missile in the sense that it minimizes the control efforts, and it possesses the desirable property that if there is no guidance system lag, the guidance command decreases to zero as the missile approaches the target. Despite its optimality and desirable property, PNG does not control the impact time explicitly. In general, the impact time required for simultaneous attacks will not be the same as the impact time of PNG. However, the introduction of an impact time error defined as the difference of these two impact times helps us derive an impact-time-control guidance law in a simple form. This study claims that the combination of a feedback loop of the impact time error with the traditional optimal feedback loop for reducing the miss-distance and minimizing the control efforts enables us to control the impact time in flight. It will be shown that the proposed law has a form of $a_c = NV\dot{\lambda} + K_\varepsilon \varepsilon_T$, where N is the navigation constant, V the missile speed, $\dot{\lambda}$ the line-of-sight rate, and ε_T the impact time error. One candidate of the feedback gain (K_ε) of the impact time error will be suggested. The new guidance law,

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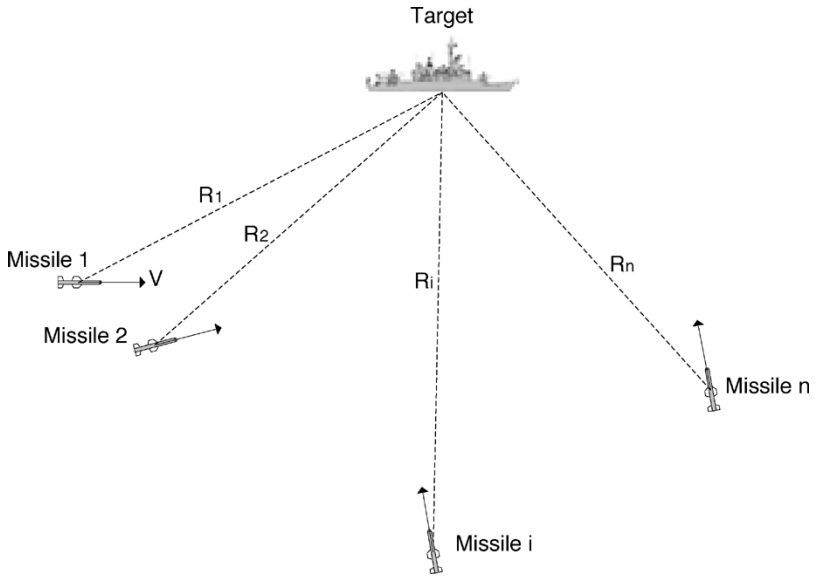


Fig. 1. An example of salvo attack.

which will be called ITCG (Impact-Time-Control Guidance) in this paper, is suboptimal in the sense that when the impact time error is reduced in flight, ITCG becomes a conventional PNG law.

This paper is organized as follows: In Section II, the impact time control problem is formulated. In Section III, the problem is solved by the maximum principle and the similarity of the closed form solution and APNG is investigated. Section IV explains the way to determine the additional command and discusses the main results. The results of nonlinear simulation are shown in Section V. Finally, the concluding remarks are given in Section VI.

II. Problem Formulation

Suppose that n missiles participate in a salvo attack against a single target as shown in Fig. 1. Although each missile has a different missile-to-target range and an initial heading angle, their common aim is to reach the target simultaneously. Two prerequisites should be mentioned here for realistic salvo attack situations: The first one is that the discrepancies in the impact times by PNG are not so much conspicuous. In other words, only the missiles with the similar ranges from the target are entitled to participate in the salvo attack. The second is that the designated impact time (T_d) for salvo attack is determined as

$$T_d \geq \max_i \{\hat{T}_i\} \text{ for any } i \in \{1, 2, \ldots, n\} \tag{1}$$

where n is the number of missiles involved in the salvo attack and $\hat{T}_i$ the estimated impact time of the i-th missile produced by PNG. It is reasonable to choose T_d to be equal or larger than the maximum of $\{\hat{T}_i\text{'s}\}$, since some missiles may not satisfy the impact time requirement otherwise.

Consider the homing guidance geometry of one of the missiles against the single target as shown in Fig. 2. The target is modeled as being stationary since the maneuverability and the speed of surface ships are not comparable with those of anti-ship missiles of high-subsonic or supersonic speed. And it is assumed

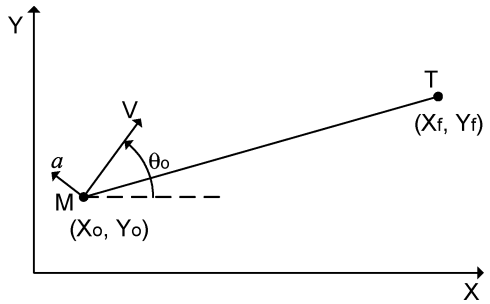


Fig. 2. Homing guidance geometry.

that the missile speed V is constant through the engagement and the autopilot lag is negligible since the missile time constant is small compared to the target time constant. In Fig. 2, X, Y, and θ denote missile positions and the heading angle in the inertial reference frame. The subscript O and f represent the initial and the final time, respectively. Assuming that the terminal time T_f is designated as T_d in (1), the governing equations in this homing problem are given by

$$\begin{aligned} \dot{X} &= V\cos\theta, \quad X(0) = X_0, \quad X(T_f) = X_f \\ \dot{Y} &= V\sin\theta, \quad Y(0) = Y_0, \quad Y(T_f) = Y_f \\ \dot{\theta} &= \frac{a}{V} = \frac{(a_B + a_F)}{V}, \quad \theta(0) = \theta_0 \end{aligned} \tag{2}$$

The missile is controlled by the acceleration command, a, which is perpendicular to missile velocity. The acceleration command is composed of two different commands: the first one is a feedback command a_B for reducing the miss-distance, and the second one is an additional command a_F for adjusting the impact time.

First, we solve an optimal control problem in which the control effort

$$J = \frac{1}{2}\int_0^{T_f} a_B^2 dt \tag{3}$$

is minimized subject to (2) under the assumption that a_F is constant. Note that this solution does not assure us of minimizing the energy of the real acceleration control (a, or $a_B + a_F$) applied to the missile. And then, we obtain the additional command, a_F, which makes the length of homing route equal to the required range-to-go, i.e., the multiple of the missile speed and the designated time-to-go.

III. Solution of Optimal Problem for Feedback Command

The linearized state equations can be written as

$$\begin{bmatrix} \frac{dy}{dx} \\ \frac{d\theta}{dx} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y \\ \theta \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_B + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_F \tag{4}$$

where the downrange x is the independent parameter, and θ is assumed to be small. Each variable is nondimensionalized as

$$t = \frac{T}{T_f}, \quad x = \frac{X}{(VT_f)}, \quad y = \frac{Y}{(VT_f)},$$
$$u_B = \frac{a_B}{\left(\frac{V}{T_f}\right)}, \quad u_F = \frac{a_F}{\left(\frac{V}{T_f}\right)}$$

An equivalent cost of (3) is given by

$$J' = \frac{1}{2}\int_{x_0}^{x_f} u_B^2(x)dx \tag{5}$$

Boundary conditions of each state variables are also nondimensionalized as

$$x(0) = \frac{X_0}{(VT_f)} = x_0, \quad x(1) = \frac{X_f}{(VT_f)} = x_f$$
$$y(x_0) = \frac{Y_0}{(VT_f)} = y_0, \quad y(x_f) = \frac{Y_f}{(VT_f)} = y_f$$

Since V is constant, the constraint on impact time can be expressed as a path constraint with respect to downrange as follows:

$$\int_{x_0}^{x_f} \sqrt{1+\theta^2(x)}dx = 1 \tag{6}$$

This linear optimization problem can be solved analytically by applying the maximum principle [12]. Let us write the Hamiltonian H for the problem as

$$H = \frac{1}{2}u_B^2 + \lambda_y\theta + \lambda_\theta(u_B + u_F) \tag{7}$$

where λ_y and λ_θ are the co-states. The optimality condition $\partial H/\partial u_B = 0$ gives the optimal acceleration command

$$u_B = -\lambda_\theta = -\nu_y(x_f - x) \tag{8}$$

where ν_y is a constant to be determined to satisfy the final conditions [12]. Integration of (4) leads to

$$\nu_y = -\frac{3\left[y_f - \theta_0 x_f - \frac{1}{2}u_F x_f^2\right)\right]}{x_f^3} \tag{9}$$

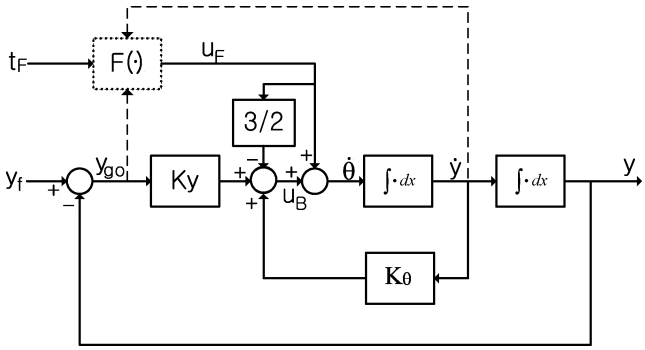


Fig. 3. Guidance loop.

Hence, the feedback guidance command at the initial time can be expressed as a function of u_F:

$$u_B = 3\left[\frac{y_f}{x_f^2} - \frac{\theta_0}{x_f}\right] - \frac{3}{2}u_F \tag{10}$$

The continuous-time feedback law is then obtained as

$$u_B = u_P - \frac{3}{2}u_F \tag{11}$$

where $u_P \equiv 3(y_{go} - \theta x_{go})/x_{go}^2$, $x_{go} \equiv x_f - x$, and $y_{go} \equiv y_f - y$. Note that u_P of (11) is the linear approximation of PNG with the navigation constant of 3. The proposed guidance loop is illustrated in Fig. 3. The block of $\mathrm{F}(\cdot)$ yielding the additional command is explained in the following section. Two guidance gains in Fig. 3 are $K_y = 3/x_{go}^2$ and $K_\theta = -3/x_{go}$.

Let us examine the characteristics of (11). In cases where the target maneuvers are significant, an extension of PNG such as augmented proportional navigation can be used, for which the commanded interceptor's acceleration depends on the target acceleration:

$$u = NV\dot{\lambda} + \frac{N}{2}u_T \tag{12}$$

where $\dot{\lambda}$ denotes the LOS rate and u_T the target maneuver acceleration [5]. Comparing the solution of (11) with APNG leads to the fact that ITCG has the form of APNG with $N = 3$.

IV. Determination of the Additional Command

A. ITCG Based on Linearized Formulation

How to determine the additional command u_F to meet the impact time requirement will be explained in this section. We start from the condition of (6), which says that the range-to-go has to be equal to the multiple of the constant missile speed and the time-to-go specified by T_d, which is the difference between the designated impact time and the current time. Note that the impact time condition needs to be satisfied only at the terminal time, but not necessarily during an engagement. This study, however, treats this condition as a constraint which should always be satisfied during the engagement. This approach has an advantage that efforts to adjust the impact time begin from the initial stage.

The heading angle, $\theta(x)$ can be expressed as a polynomial of x from (4) and (11)

$$\theta(\eta) = \alpha\eta^2 + \beta\eta + \gamma \tag{13}$$

for $\eta \in [x, x_f]$, where

$$\begin{aligned}
\alpha &= \frac{\nu_y}{2} \\
\beta &= u_F - \nu_y x_f \\
\gamma &= \theta(x) + \nu_y x_f x - \frac{\nu_y x^2}{2} - u_F x \\
\nu_y &= -\frac{3\left[y_f - \theta(x)x_f - \frac{1}{2}u_F x_f^2\right)\right]}{x_f^3}
\end{aligned}$$

Taylor series expansion of (6) over θ with the higher order terms neglected yields

$$\begin{aligned}
\bar{t}_{go} &= \int_x^{x_f} \sqrt{1+\theta^2(\eta)}d\eta \approx \int_x^{x_f} \left(1 + \frac{1}{2}\theta^2(\eta)\right) d\eta \\
&= x_{go} + \frac{1}{2}\sum_{n=1}^{5} \frac{c_n}{n}\left(x_f^n - x^n\right)
\end{aligned} \tag{14}$$

where $\bar{t}_{go}$ is the designated time-to-go and c_n's are the coefficients of polynomial of $\theta^2(x)$ given as

$$\begin{aligned}
c_1 &= \frac{\gamma^2}{2} \\
c_2 &= \beta\gamma \\
c_3 &= (\beta^2 + 2\alpha\gamma) \\
c_4 &= \alpha\beta \\
c_5 &= \frac{\alpha^2}{2}
\end{aligned}$$

This equation is expressed in terms of u_F as

$$u_F^2 + 2u_P u_F + 16u_P^2 + \frac{80\theta u_P}{x_{go}} + \frac{240\left(1 + \frac{\theta^2}{2}\right)}{x_{go}^2} - \frac{240\bar{t}_{go}}{x_{go}^3} = 0 \tag{15}$$

Letting u_F be zero in (15) results in the useful approximation of the time-to-go of PNG, $\hat{t}_{go}$, as follows:

$$\hat{t}_{go} = \frac{1}{15}u_P^2 x_{go}^3 + \frac{1}{3}u_P\theta x_{go}^2 + \left(1 + \frac{\theta^2}{2}\right)x_{go} \tag{16}$$

The solutions of (15) are given by

$$u_F = -u_P \pm \sqrt{u_P^2 + \frac{240}{x_{go}^3}\varepsilon_t} \tag{17}$$

where ε_t is an impact time error, i.e., $\varepsilon_t = \bar{t}_{go} - \hat{t}_{go}$. Choosing the solution satisfying the condition that $u_F = 0$ if $\varepsilon_t = 0$, we obtain the additional command as

$$u_F = -u_P\left(1 - \sqrt{1 + \frac{240}{u_P^2 x_{go}^3}\varepsilon_t}\right) \tag{18}$$

A positive impact-time error guarantees the existence of a u_F. A negative impact-time error implies that the missile must home to the target along a shorter route than the PNG trajectory. This flight is, however, not always possible but only for the range of $\varepsilon_t \geq -x_{go}^3 u_P^2/240$. Hence if $u_P = 0$, i.e., $\dot{\lambda} = 0$, then ε_t should be nonnegative. If $u_P \neq 0$, then ε_t is allowed to have some negative value. The RHS of this inequality can be considered as *the margin* of the impact-time error. Now we examine the meaning of this condition. Without loss of generality, y_{go} can be set to be zero at initial time. Observe that there exists an u_F satisfying (17) if $\varepsilon_t \geq -(3/80)\theta^2 x_{go}$ or $\bar{t}_{go} \geq (1 + \theta^2/16)x_{go}$. In addition, if θ is zero, this condition is always satisfied whenever the designated time-to-go is greater than the flight time along the straight course from the missile to the target.

Combining (11) with (18) yields the acceleration command, which is the proposed ITCG law based on the linear formulation, as follows:

$$\begin{aligned}
u_B + u_F &= u_P - \frac{1}{2}u_F \\
&= u_P + \frac{1}{2}u_P\left(1 - \sqrt{1 + \frac{240}{u_P^2 x_{go}^3}\varepsilon_t}\right) \\
&= u_P\left(\frac{3}{2} - \frac{1}{2}\sqrt{1 + \frac{240}{u_P^2 x_{go}^3}\varepsilon_t}\right)
\end{aligned} \tag{19}$$

As the missile approaches the target, the control command may blow up as (19) implies that the sensitivity of u_F to the time-to-go error is inversely proportional to x_{go}. To circumvent this undesirable property of ITCG, a simple switching rule is proposed as follows: ITCG is switched to PNG once the calculated impact time error is reduced below a certain small value. A small impact time error means that the range-to-go by PNG is approximately equal to that of ITCG. So conversion to PNG yields smooth homing with acceptable impact time errors.

B. Discussion

The expression on ITCG with physical units corresponding to (19) for nonlinear simulation in the time domain is as follows:

$$a = a_B + a_F = NV\dot{\lambda}\left(\frac{3}{2} - \frac{1}{2}\sqrt{1 + \frac{240V^5}{(NV\dot{\lambda})^2 R_{go}^3}\varepsilon_T}\right) \tag{20}$$

where $\varepsilon_T = \bar{T}_{go} - \hat{T}_{go}$ and $R_{go} = \sqrt{X_{go}^2 + Y_{go}^2}$. Furthermore, as the expression of (16) in a polar frame along LOS can be rewritten as

$$\hat{t}_{go} \approx \left(1 + \frac{(\theta - \lambda)^2}{10}\right) r_{go} \tag{21}$$

where $r_{go} = \sqrt{x_{go}^2 + y_{go}^2}$ and λ is a LOS angle, the time-to-go estimation with physical units can be expressed as

$$\hat{T}_{go} = \frac{\left(1 + \frac{(\theta-\lambda)^2}{10}\right) R_{go}}{V} \tag{22}$$

Since this approximation becomes accurate when the angle between the missile velocity and LOS is reduced, the accuracy of estimation increases as a missile approaches to a target.

TABLE I
PARAMETER VALUES USED IN SIMULATION

Parameters	Values
Initial Missile Position	(0, 0) m
Missile Velocity	300 m/s
Target Position	(10000, 0)m
Initial Heading Angle	30 deg
Designated Impact Time	40 sec

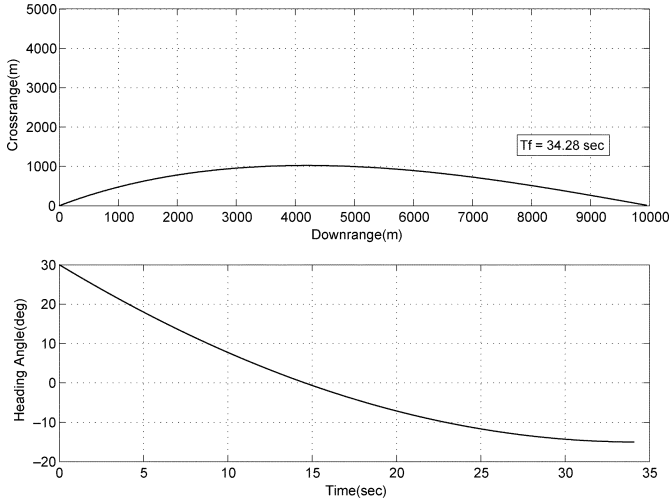


Fig. 4. Trajectories by PNG.

The square root term of (19) can be approximated into the first order term of the impact time error when the impact time error is not large as follows:

$$\sqrt{1+\frac{240\varepsilon_t}{u_P^2 x_{go}^3}} \approx 1+\frac{120\varepsilon_t}{u_P^2 x_{go}^3} \tag{23}$$

Hence, (20) is expressed as

$$a = a_B + a_F = NV\dot{\lambda} + K_\epsilon \varepsilon_T \tag{24}$$

where $K_\epsilon = -120V^5/(NV\dot{\lambda}R_{go}^3)$. This approximation shows the simple structure, the combination of the well-known PNG and the feedback of the impact time error, hidden in the suggested law of (20). Notice that the proposed law turns out to be PNG if the impact time error, ε_T, is zero. And the impact time error means a difference between the desired impact time and the expected impact time when the missile keeps PNG-homing without any efforts to adjust the miss in impact time.

The nonlinear simulation using (2), (22), and (20) [or (24)] will show that adding a feedback loop of the impact time error to PNG enables us to control the impact time.

V. NUMERICAL EXAMPLES

A. Performance of ITCG

In order to investigate the characteristics of the proposed law, nonlinear simulations with the boundary conditions shown in Table I are performed. In the case of PNG, the impact time is observed in simulation to be about 34.28 sec as shown in Fig. 4, which is almost the same as the approximated impact time (34.25 sec) of (22). Note that this is quite different from the designated impact time of 40 sec.

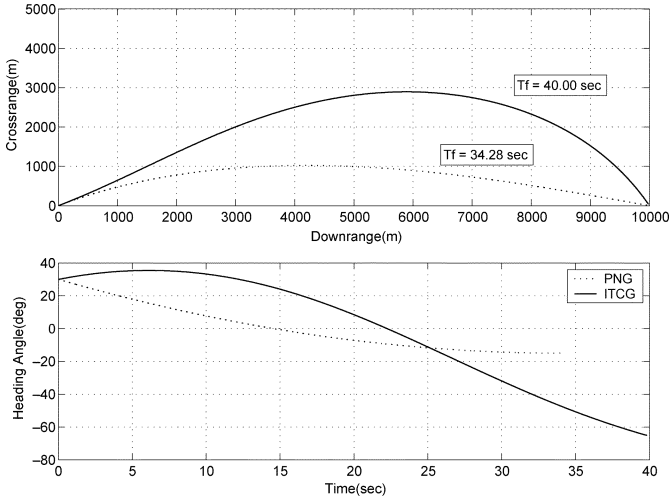


Fig. 5. Trajectories by ITCG.

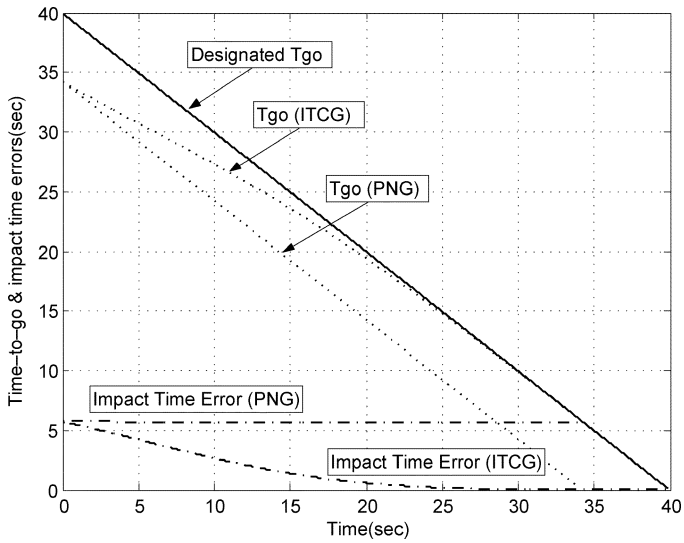


Fig. 6. Time-to-go and impact time errors.

In the case of ITCG, however, the impact time is observed to be about 40.0 sec, quite equal to the desired impact time. The solid-line in Fig. 5 depicts the trajectory by ITCG, producing much longer route than PNG. Fig. 5 also compares the heading angle profiles of PNG and ITCG. The heading angle increases at the beginning and then decreases for ITCG, while it decreases monotonically for PNG. Fig. 6 shows clearly that the impact-time error decreases from the early stage with ITCG while it does not with PNG. As a result, it is observed in Fig. 7 that ITCG employs more control energy than PNG.

Fig. 8 shows the trajectories for the proposed law with different designated impact times such as 36, 39, 42, and 45 sec. The other parameters are the same as in Table I. The impact time errors are within 0.1 sec for all cases. Fig. 9 shows the trajectories and heading angles for various initial heading angles of 10, 20, 30, 40, and 50 deg, respectively. As expected, ITCG yields

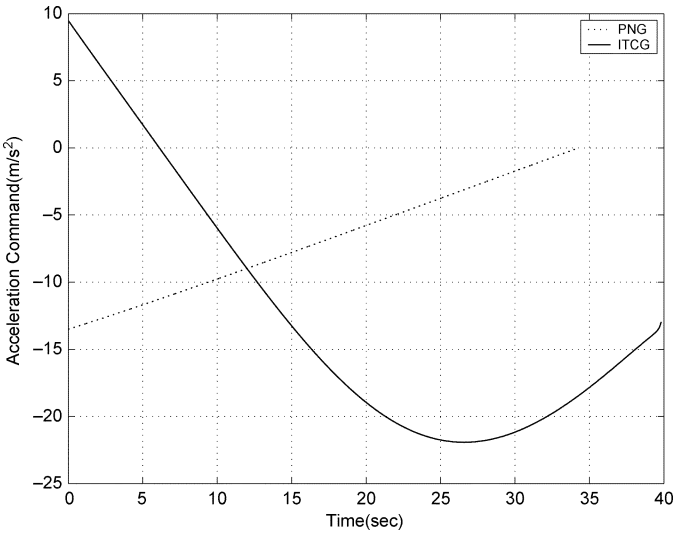


Fig. 7. Acceleration commands.

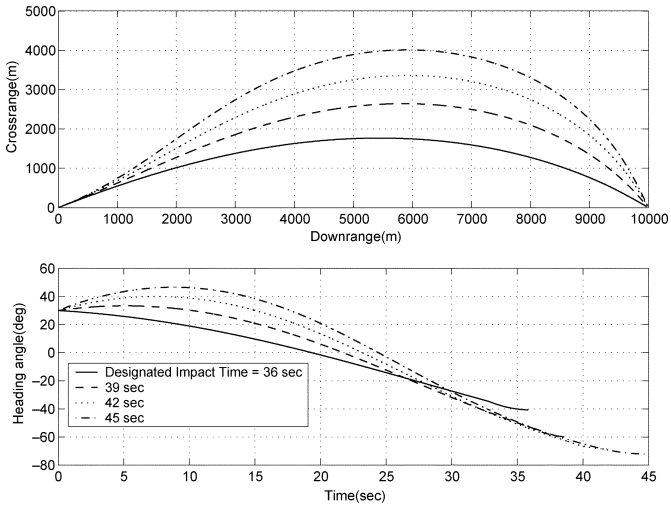


Fig. 8. Trajectories with different designated impact times.

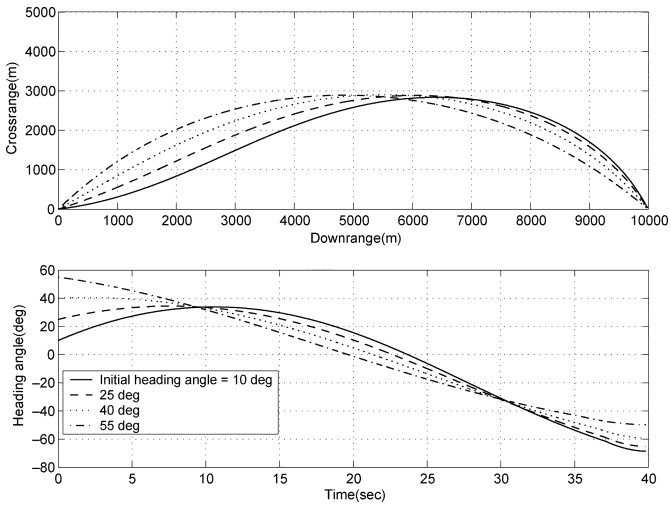


Fig. 9. Trajectories with different initial heading angles.

almost exactly 40.0 sec in flight time regardless of the initial heading angle.

TABLE II
SCENARIO FOR SALVO ATTACK

	Parameters	Values
Target	Position	(0, 0)m
Missile 1	Position	(-9093, 5250) m
	Heading angle	-35 deg
Missile 2	Position	(-6364, -6364) m
	Heading angle	75 deg
Missile 3	Position	(-2070, -7727) m
	Heading angle	105 deg
Missile 4	Position	(4750, -8227) m
	Heading angle	105 deg

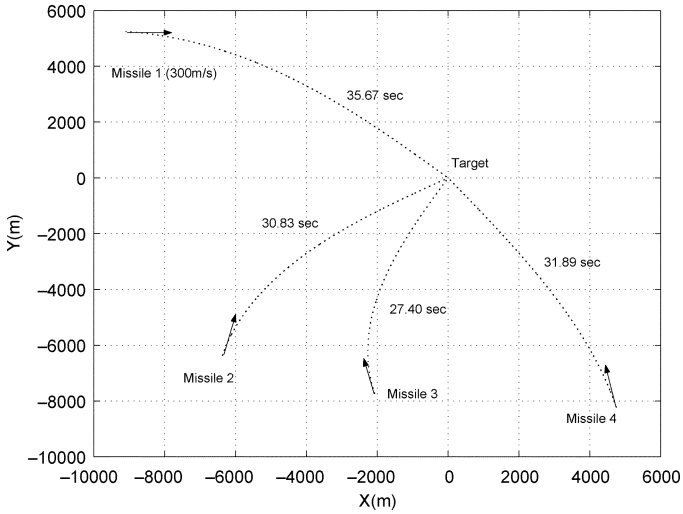


Fig. 10. Salvo attack by PNG.

B. Application of ITCG to a Salvo Attack Scenario

Suppose that four missiles attack a single target with the initial conditions shown in Table II. Each missile has the same speed, 300 m/s, and the designated impact time is determined as 37 sec, which is selected to be slightly larger than the maximum of estimated impact times by PNG.

Fig. 10 illustrates the salvo attack trajectories of four missiles by PNG and shows large discrepancies in the impact time of each missile. In this scenario, it is observed that the dispersion of impact time is about 8.3 seconds. On the other hand, Fig. 11 explicitly demonstrates that ITCG can drive the four missiles to hit the target nearly simultaneously at the desired impact time. In this example, ITCG reduces the impact time dispersion within about 0.1 seconds around the designated value. Even though the maneuverability of surface ships during the missile homing is not comparable with that of anti-ship missiles of high-subsonic or supersonic speed as previously mentioned, maneuvering of a surface target will bring about some error in impact time. The worst case on an impact time error is the case when the target moves along or against the incoming missile direction. It is investigated in this scenario that if the target moves with 45 degree heading angle in speed of 20 m/s, it causes about 1.2 seconds dispersion in impact time.

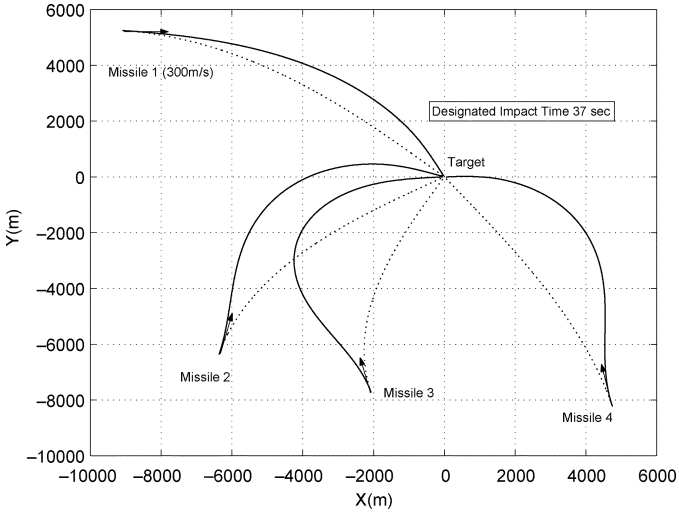


Fig. 11. Salvo attack by ITCG.

VI. Concluding Remarks

The guidance problem with impact time constraints, which has not been treated in the open literature so far, is studied in this paper. A useful suboptimal feedback control law called ITCG, which has a form of the combination of the well-known PNG and the feedback of the impact time error, is derived. The proposed law has a simple form, which leads to an easy implementation. The performance and the feasibility of the law are demonstrated through nonlinear simulations of various engagements against stationary or slightly moving targets. The proposed law provides satisfactory performance for salvo attacks of anti-ship missiles in terms of impact time and miss-distance. In future study several factors such as the heading angle constraints by the seeker gimbal limit or the final impact angle constraints should be considered for practical implementations.

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